AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training and Entrepreneurial Development

Hello!!!

Silence Hour

Use this time to revise your notes before we begin.

AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training and Entrepreneurial Development

ICT233

Web Technologies and Design.

Lecture Four

Today's Lecture

1.0 Formatting HTML Documents

2.0 Creating Lists of Items in HTML

3.0 Linking web pages in HTML

4.0 Adding Multimedia to web pages in HTML

5.0 * You have 5 key Activities to do. Prepare

Lecture Objectives

- By the end of the lecture, the student will be able to:
  - Design web pages using varied HTML tags
  - Add list of items to a web page using HTML
  - Design web pages and link them together
  - Add multimedia contents to web pages
  - Present tabular reports on a web page

Formatting HTML Documents

- Formatting is about modifying the appearance of text and other contents on a web page.
- According to the current trend of web development, formatting web pages is better handled using CSS.
- The commonly used tags for formatting within HTML documents are:
- **<h1>** Heading 1**</h1>.** The heading tags are from **<h1>** through to **<h6>**, with **<h6>** being the smallest.
- ** **Paragraph** <pre>** Preformatted Text**</pre>**

HTML Text Formatting Tags

Tag	Description
<b>	Defines bold text
	Defines emphasized text
<i>	Defines italic text
<small>	Defines smaller text
	Defines important text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<ins>	Defines inserted text
<del>	Defines deleted text
<mark>	Defines marked/highlighted text

Formatting HTML Documents

- Note: Browsers display **** as **<b>**, and **** as **<i>**.
- However, there is a difference in the meaning of these tags: <b>and <i> defines bold and italic text,
  but and means that the text is "important".

HTML Quotation and Citation Tags

Tag	Description
<abbr>	Defines an abbreviation or acronym
<address>	Defines contact information for the author/owner of a document
<bdo>	Defines the text direction
<blockquote>	Defines a section that is quoted from another source
<cite>	Defines the title of a work
<q>	Defines a short inline quotation

The HTML <abbr> Element

- The HTML <abbr> element defines an abbreviation or an acronym.
- Marking abbreviations can give useful information to browsers, translation systems and search-engines.
- ** The <abbr title="World Health Organization">WHO</abbr> was founded in 1948. **

The HTML <q> for Short Quotations

- The HTML **<q>** element defines a short quotation.

** **Check this out:

<q>Ignore the distractions around you, for in the calmness of your mind , you will experience its enormous capabilities**.</q>**

** **

The HTML <cite> for Work Title

- The HTML <cite> element defines the title of a work.
- Browsers usually display <cite> elements in italic.
- E.g. ** <cite>**The Scream**</cite>** by Edvard Munch. Painted in 1893**. **

The HTML <bdo> Element

- The HTML <bdo> element defines **bi-directional override**.
- The <bdo> element is used to override the current text direction:
- **E.g. <bdo dir="rtl">This text will be written from right to left</bdo>** when previewed in the web browser produces**:**

tfel ot thgir morf nettirw eb lliw txet sihT

Activity One – 15 Minutes

1. Create a folder named **ICT233--2026** at a preferred location on your computer.
2. Inside your **ICT233--2026** folder**,** create sub folders named: **multimedia**, **css**, and **js.**
3. Launch your preferred text editor, and setup a complete HTML page using the HTML5 document structure we discussed in our previous lecture.
4. Make use of all the HTML elements we have discussed so far in today's lecture to add your own contents to your HTML page.
5. Save the page as **index.html** in the **ICT233-2026** folder and preview it in your chosen web browser.

Activity Two – 5 Minutes

1. Write down the list of all the courses you have registered for this semester
2. Make a list of 5 movies or TV Shows you have watched in recent times.
3. For each of the movies listed, write down at least 3 characters.

Creating Lists in HTML

1. HTML supports **ordered**, **unordered** and **definition** lists.
2. It is also possible to nest one type of list within another in HTML documents.

Ordered List

- An ordered list is a list in which the list items are numbered sequentially.
- Ordered lists are enclosed in the **<ol> </ol>** tags.
- Each list item is placed within the **<ol> </ol>** tags using the **<li> </li>** tags
- Example:

```
<h3>my favorite reggae icons:</h3>
    <ol type="A" >      <!-- <ol style="list-style-type: upper-alpha;"> -->
        <li> Joseph Hill </li>
        <li> Burning Spear</li>
        <li> Buju Banton </li>
        <li> Bob Marley </li>
    </ol>
```

The *type* attribute of the <ol> tag.

- You can display different kinds of ordered list using the type attribute of the **<ol>** tag .
- The table below is a guide to using the type and style attributes.

TYPE	STYLE	DESCRIPTION	EXAMPLES
A	upper-alpha	Uppercase letters	A. B. C. D.
a	lower-alpha	Lowercase letters	a. b. c. d.
I	upper-roman	Uppercase Roman numerals	I. II. II. IV
i	lower-roman	Lowercase Roman numerals	i. ii. Iii. iv
1	decimal	Arabic numerals(default)	1. 2. 3. 4.

Unordered List - 1

- Unordered list is used when the list of items does not have any necessary sequence of appearance.
- It is also called a ***bulleted list***.
- A bullet is a decorative label assigned to each item in a list.
- Unordered lists are enclosed in the **<ul>** **</ul>** tags.
- Each list item is placed within the **<ul>** **</ul>** tags using the **<li>** **</li>** tags
- The syntax for the unordered list is similar to that of ordered list with the type attribute set to ***circle, disc*** or ***square.***
- You may also use the ***style attribute*** to specify your type of bullet

Unordered List - 1

- A **style** attribute can be added to an **unordered list**, to define the style of the marker:

Style	Description
list-style-type:disc	The list items will be marked with bullets (default)
list-style-type:circle	The list items will be marked with circles
list-style-type:square	The list items will be marked with squares
list-style-type:none	The list items will not be marked

Unordered List - Example

```
<html>
    <head>
        <title>Unordered List </title>
    </head>
    <body>
            The following are my favorite reggae icons:
            <ul type="square" >   <!-- <ul style="list-style-type: square;"> -->
                <li> Joseph Hill </li>
                <li> Burning Spear</li>
                <li> Buju Banton </li>
                <li> Bob Marley </li>
            </ul>
    </body>
</html>
```

Definition Lists

- This is used to present a term and its definition formatted like a glossary or a dictionary.
- It is the ideal format to present lists of words or phrases with their meanings.
- The tags used are

```
<dl>
    <dt> Definition Term </dt>
    <dd> Definition Description </dd>
</dl>
```


Summary of List Elements

Tag	Description
<ul> </ul>	Defines an unordered list
<ol> </ol>	Defines an ordered list
<li> </li>	Defines a list item
<dl> </dl>	Defines a description list
<dt> </dt>	Defines the term in a description list
<dd> </dd>	Defines the description in a definition list

Activity Three – 10 Minutes

1. Based on the two previous activities,
2. Create a new web page named **lists.html**
3. On this new page, reproduce the list you created in **Activity Two.**

File Paths

- A file path describes the location of a resource in a web site's folder structure.
- File paths are used when locating files and web resources such as:
  - Web pages
  - Multimedia contents (images, video, and audio)
  - Style sheets
  - JavaScripts

File Paths – Absolute and Relative

- An absolute file path is the full URL to an internal or external resource:

```
<img src="https://www.w3schools.com/images/picture.jpg" alt="Mountain">
```

- A relative file path points to a resource relative to the current page.
- In this example, the file path points to a file in the images folder located at the root of the current web page:

```
<img src="images/myPicture.jpg" alt="Mountain">
```


Creating Links- 1

- A static web page contains no links within the document or to other documents.
- In order to give users some interactivity and control over our web pages, we need to incorporate links to files or other pages of interest.
- Creating links in HTML documents is an important step in web site development because it adds the "hyper" to the term hypertext.

Creating Links - 2

- A hypertext document is one that contains elements called **LINKS** that allow users to activate a particular part of the screen and perform some action.
- These actions include:
  1. Moving to another part of the same web page.
  2. Opening another web page on the same web site.
  3. Opening a web page on a web site somewhere else in the world.

Types of Links

- There are three main types of links namely:

1. Intra-page links

– These are links to items located within a single document or web page.

2. Intra-system links

– These are links to other pages of the same web site.

3. Inter-system links

– These are links to web pages or documents located on other web sites or web servers.

HTML Link Syntax

- The general format for creating links in HTML is given by:

<a href="URL">Object to click</a>

- The URL defines the path to the resource to be displayed when one clicks on the "**Object to click**".
- The "**Object to click**" defines the part to be displayed and it does not have to be always text.
- It could also be an image or any other HTML element.
- The **href** (Hypertext Reference) attribute defines the link's address.

The target Attribute

- The target attribute of the **<a>** tag defines where the linked document will be opened.
- The code below opens the linked document in a new browser window.

```
<html>
    <head> <title>Setting the target attribute to _blank </title> </head>
  <body>
<a  href="http://www.aamusted.edu.gh/"  target="_blank"> Visit
   AAMUSTED</a>
    <p>
       If you set the target attribute of the <a> tag  to "_blank", the
       document will open in a new browser window.  </p>
    </body>
</html>
```

Values for the target Attribute.

ATTRIBUTE VALUE	DESCRIPTION
_blank	Opens the linked document in a new browser window.
_self	Opens the linked document in the same frame as it was clicked(this is default).
_parent	Opens the linked document in the parent frameset.
_top	Opens the linked document in the full body of the browser window.
_framename	Opens the linked document in a named frame

Links – Creating a Bookmark

- HTML bookmarks are used to allow readers to jump to specific parts of a Web page.
- Bookmarks can be useful if your webpage is very long.
- To make a bookmark, you must first create the bookmark, and then add a link to it.
- When the link is clicked, the page will scroll to the location with the bookmark.
- This is also referred to as **named anchor**.

Links – Creating a Bookmark

- First, create a bookmark with the **id** attribute:

<h2 id="C4">Chapter 4</h2>

- Then, add a link to the bookmark ("Jump to Chapter 4"), from within the same page:

<a href="#C4">Jump to Chapter 4</a>

- Or, add a link to the bookmark ("Jump to Chapter 4"), from another page:
- Example

<a href="html_demo.html#C4">Jump to Chapter 4</a>

Some Examples of Links

- Intra-Page Link:

<a href="#L5">Lecture Five</a>

- Intra-System Link:

<a href="contacts.html">Contact Us</a>

- Inter-System Link:

<a href="https://www.google.com/">Google</a>

The URL for intra-page links use named anchors.

The URL for intra-system links is mostly relative.

The URL for inter-system links is absolute

Linking to Email Addresses

- In addition to linking between parts of a single web page, web pages and websites, the <a> tag allows you to link to email addresses.
- This is the simplest way to enable visitors to your site, to talk back to you via a clickable link to your email address.
- An HTML link to an email address looks like the following:

<a href="mailto:yourusername@yourdomain.com">Email me.</a>

Example: **<a href="mailto:jmarvin@wavetek.com">Email us.</a>**

Activity Four – 15 Minutes

1. Based on the pages you created in the earlier activities, link the pages such that, when we are on one page, we should have a link that would redirect us to the other page.
2. Add a link on each of the pages that would redirect as to AAMUSTED's website (https://www.aamusted.edu.gh), and Google's website (https://www.google.com)
3. Remember to open the websites of AAMUSTED and Google in their own tabs.

Adding Multimedia in HTML

- Without multimedia in HTML documents, a web page is just a collection of text.
- Multimedia is collective name for images, music, sound, video, animations and other special effects on a web page.
- The support for multimedia is handled in different ways by different browsers.
- Some multimedia elements can be handled inline, and some require an extra helper program referred to as ***plug-in***.

Common Image Formats

IMAGE FORMAT	DESCRIPTION	FILE EXTENSION
GIF	Graphics Interchange Format	.gif
JPEG	Joint Photographers Experts Group	.jpg/.jpeg
PNG	Portable Network Graphics	.png
PDF	Portable Document Format	.pdf
TIFF	Tagged Image File Format	.tif/.tiff
BMP	Bit Mapped Picture	.bmp

Adding Images to Web Pages

- Images are inserted into HTML documents using the the ***<img />*** tag.
- The ***src*** attribute is the only required attribute of the ***<img />*** tag.
- This attribute tells the browser where to find the image that is to be inserted into the web page.
- The ***alt*** attribute provides a text-based description of the image.
- Other attributes of the ***<img />*** tag include ***height, width and align***

Example with the <img /> tag

```
<html>
    <head>
        <title>Adding Images</title>
    </head>
    <body style="background-color:#fff;">   <!-- Background color is
      set to white -->
        <p>Here is a picture of a great reggae icon </p>
        <img src="images/culture.jpg" alt= "Joseph Hill" />
          <br>
        <p>Actually , this is  Joseph Hill (Culture) from
          Jamaica. </p>
    </body>
</html>
```

Adding Audio Content to Web Pages

- The **<audio>** tag is used to embed sound content such as music or voice streams in web pages .
- The **<audio>** tag contains one or more **<source>** tags with different audio sources.
- The web browser chooses the first source it supports.
- The text between the **<audio>** and **</audio>** tags will be displayed for browsers that do not support the **<audio>** element.
- The three highly supported audio formats in HTML are **MP3**, **WAV**, and **OGG**.

Adding Audio Content - Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Audio Element</title>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
<h2>Adding Audio Content</h2>
<p>Click on the play button to play a sound:</p>
<audio controls>
  <source src="sounds/coolsound.ogg" type="audio/ogg">
  <source src="sounds/coolsound.mp3" type="audio/mpeg">
  Your browser does not support the audio element, update it.
</audio>
</body>
</html>
```

Adding Video Content to Web Pages

- Video files are collection of images usually with a related sound file.
- The **<video>** tag is used to embed video content such as a movie clip or other video streams in web pages .
- The **<video>** tag contains one or more **<source>** tags with different video sources.
- The web browser chooses the first source it supports.
- The text between the **<video>** and **</video>** tags will be displayed for browsers that do not support the **<video>** element.
- The three highly supported video formats in HTML are **MP4**, **WebM**, and **OGG**.

Adding Video Content - Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Video Element</title>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
<h2>Adding Video Content</h2>
<p>Click on the play button to play the video:</p>
<video width="320" height="240" controls>
<source src="videos/coolvideo.mp4" type="video/mp4">
<source src="videos/coolvideo.webm" type="video/webm">
<source src="videos/coolvideo.ogg" type="video/ogg">
Your browser does not support the video element.
</video>
</body>
</html>
```

Activity Five – 15 Minutes

- Visit https://www.unsplash.com
- Search for free images and download a minimum of five images of your choice and save them with appropriate names in your **multimedia** sub folder, you created in **Activity One**.
- Using the **<img />** element, add at least two of the images to your html page.
- Use the **<video> </video>** and **<audio> </audio>** elements to video and audio files to your HTML page.
- Now go to https://www.youtube.com
- Search for a video of your choice.
- Right click on the video and click '**Copy embed code**'
- Paste the copied code into your HTML page.
- Save the page as **multimedia.html** and preview it in your web browser.

Our Next Lecture …

1. **Presenting tabular information on web pages.**

2. **Accepting User Input with HTML Forms..**

Presenting Information in Tables

- Tables provide an excellent way to organize and display information on web pages.
- Tables are defined using the **<table> </table>** tag pair.
- A table is divided into rows with the **<tr> </tr>** tag pair, and each row is divided into data cells using the **<td> </td>** tag pair.
- The letters **td** stand for "table data" which is the content of a data cell.
- A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, and so on.

Presenting Information in Tables

- The HTML **<table> element** contains the table information, which consists of **table header elements(<th>)**, **table row elements(<tr>)**, and individual **table data cells(<td>)**.
- These are the three elements used most frequently when you are building tables.

HTML Table Elements

Element	Description
<table>…..</table>	Establishes the table. It contains all other elements that specify caption, rows, and content
<tr>…..</tr>	Table row. It contains the table cells.
<td>……</td>	Table data cell. This contains the actual table data.
<th>……</th>	Table header cell: It contains header information for a column of data.
<caption>….</caption>	Provides a short description of the table's contents.
<thead>…..<thead>	Signifies table header.
<tbody>…..<tbody>	Signifies table body
<tfoot>……<tfoot>	Signifies table footer

Using the border and width Attributes

- The border attribute is used to specify the thickness of the lines that make the table.
- The width attribute specifies how wide the table should be in the browser window or relative to other layout elements.
- It is a good practice to specify the table width in percentages.
- The example below sets values the border and the width attributes of a table.

```
<table border="1" width="35%">
        Other codes that make up the table go here
</table>
```

HTML Table Example

```
<!DOCTYPE html>
<html>
<head> <title> HTML Tables: </title> </head>
<body>
    <table  width="45%" border="1" style="border-collapse:collapse;">
    <caption> <h1>My Favorite Collections</h1> </caption>
      <tr>  <th>Title of Music </th> <th> Artiste </th>  </tr>
      <tr>  <td> Forward to Africa </td>  <td> Joseph Hill </td>  </tr>
        <tr>  <td> China Roses </td>  <td> Enya </td>   </tr>
        <tr>  <td> Floral Street </td>  <td> Enya </td> </tr>
        <tr> <td> Humble Africa </td>  <td> Joseph Hill </td> </tr>
        <tr>    <td> Nah Stay Inna Babylon</td>  <td> Joseph Hill </td> </tr>
        <tr> <td> Miss Independent </td>  <td> Neyo </td> </tr>
        <tr> <td> Storm in Africa</td> <td> Enya </td>  </tr>
        <tr> <td> Count on Me</td> <td> Whitney Houston</td> </tr>
        </table>
    </body>
</html>
```


Spanning Columns

- The **colspan** attribute lets you create cells that span multiple columns of a table.
- Column cells always span to the right.
- The syntax is given as:

```
<tr>
        <td colspan="number of columns"> Content goes here.</td>
</tr>
```

- **Example:**

```
<tr>
        <td colspan="3">
                This text spans across 3 columns.
        </td>
</tr>
```

Spanning Rows

- The **rowspan** attribute lets you create cells that span multiple rows of a table.
- Rows always span down.
- The syntax is given as:

```
<tr>
        <td rowspan="number of rows"> Content goes here.</td>
</tr>
```

- **Example:**

```
<tr>
        <td rowspan="4"> This text spans across 4 rows down.</td>
        <th>Name</th>
        <th>Phone</th>
        <th>Email</th>
</tr>
```


Organizing HTML Table Contents

- For code organization, rows can be grouped into head, body, and footer sections using the **<thead>**, **<tbody>**, and **<tfoot>** elements respectively. E.g

```
<table>
   <thead>
       <tr><th>Header Cell 1</th> <th>Header Cell 2</th> </tr>
   </thead>
   <tbody>
        <tr> <td>Body Cell 1</td> <td>Body Cell 2</td> </tr>
         <tr><td>Body Cell 3</td> <td>Body Cell 4</td> </tr>
   </tbody>
   <tfoot>
       <tr><td>Footer Cell 1</td> <td>Footer Cell 2</td> </tr>
   </tfoot>
</table>
```

Applying Style to the Table

- Styling table elements enhances the appearance of the table, and the information presented in it.
- This is best achieved using CSS.
- The CSS style rules could be specified within the head section of the HTML file using the **<style>** element.
- We provide a CSS style rule code segment that styles any given HTML table.

Sample CSS rule

```
<style type="text/css">
    table{
        border: solid 1px black;        border-collapse: collapse;
        font-family: Georgia, Serif, "Times New Roman";    color: #722750;
       }
    td, th {
        border: solid 1px black;        padding:  5px;
        }
    th   {
        font-family: "Trebuchet MS", Georgia, Serif, Verdana;    font-size:  14px;
        font-style:  bold;    background-color:  #7fa2c1;    color:  #0070dd;
       }
    td   {
        font-family: Georgia, Serif, Verdana;    font-size:  12px;
        background-color: #ccdae6;
      }
</style>
```

Creating Background Hover Effects

- Hover effects could be used to add interactivity to tables by using CSS rules.
- When a user hovers the pointer over a cell or row then the background, font, or formatting can change.
- An example is shown below:

```
td {
        background-color:  #ccdae6;
    }
td:hover  {
            color: white;
            background-color:  #722750;
          }

tr:hover  td {
              color: white;
              background-color:  #722750;
              }
```


Focus Questions?

1. Create three related web pages and link them together using the **<a> </a>** tag.
2. Using the **<img />** tag, insert your own photograph in one of the pages created above.
3. Using the <audio> and <video> tags insert audio and video contents into the web pages you created.

- Visit https://www.w3schools.com and read on the following:
    1. **HTML Layout**
    2. **HTML Responsive Web Design**
    3. **HTML5 new Elements.**

Please bring any questions you may have about these areas to class for discussion.

Our Next Lecture …

1. Accepting User Input with HTML Forms.

2. Layout Design and Responsive Web Design.

3. HTTP Methods and Messages.